

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 2

Wednesday 9 November 2022

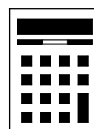
07:00 GMT

Time allowed:
1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



N 0 V 2 2 9 2 0 2 2 0 1

Answer **all** questions in the spaces provided.

Do not write
outside the
box

0 1

This question is about atoms, elements and compounds.

0 1 . 1

Which particle in the atom has the lowest mass?

[1 mark]

Tick (✓) **one** box.

Electron

☐

Neutron

☐

Proton

☐

0 1 . 2

Which particle in the atom has no charge?

[1 mark]

Tick (✓) **one** box.

Electron

☐

Neutron

☐

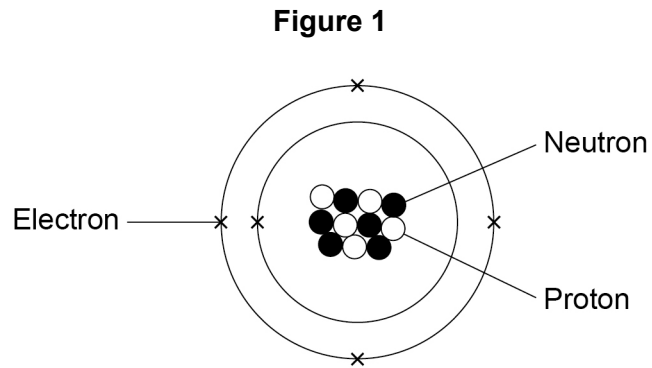
Proton

☐

0 1 3 An atom of beryllium is represented as ${}^9_4\text{Be}$

A student drew a diagram of the structure of an atom of ${}^9_4\text{Be}$

Figure 1 shows the student's diagram.



Give **four** reasons why the student's diagram is **not** correct.

[4 marks]

- 1 _____
- 2 _____
- 3 _____
- 4 _____

Question 1 continues on the next page

Turn over ►



0 1 . 4 An atom of iron is represented as $^{56}_{26}\text{Fe}$

What is the name of the part of the periodic table where iron is found?

Use the periodic table.

[1 mark]

Tick (✓) **one** box.

Alkali metals

☐

Halogens

☐

Noble gases

☐

Transition metals

☐

0 1 . 5 The formula of a compound of iron is $\text{Fe}(\text{OH})_3$

Calculate the relative formula mass (M_r) of this compound.

Relative atomic masses (A_r): H = 1 O = 16 Fe = 56

[2 marks]

Relative formula mass = _____

9



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 2

This question is about the Group 7 elements and ionic compounds.

0 2

. 1

Which of the following Group 7 elements has the highest melting point?

Use the periodic table.

[1 mark]

Tick (✓) **one** box.

Bromine

☐

Chlorine

☐

Fluorine

☐

Iodine

☐

0 2

. 2

Table 1 shows the size of the atoms of three Group 7 elements.

Table 1

Element	Size of atom in nm
Chlorine	0.10
Bromine	0.11
Iodine	0.13

Complete **Figure 2**.

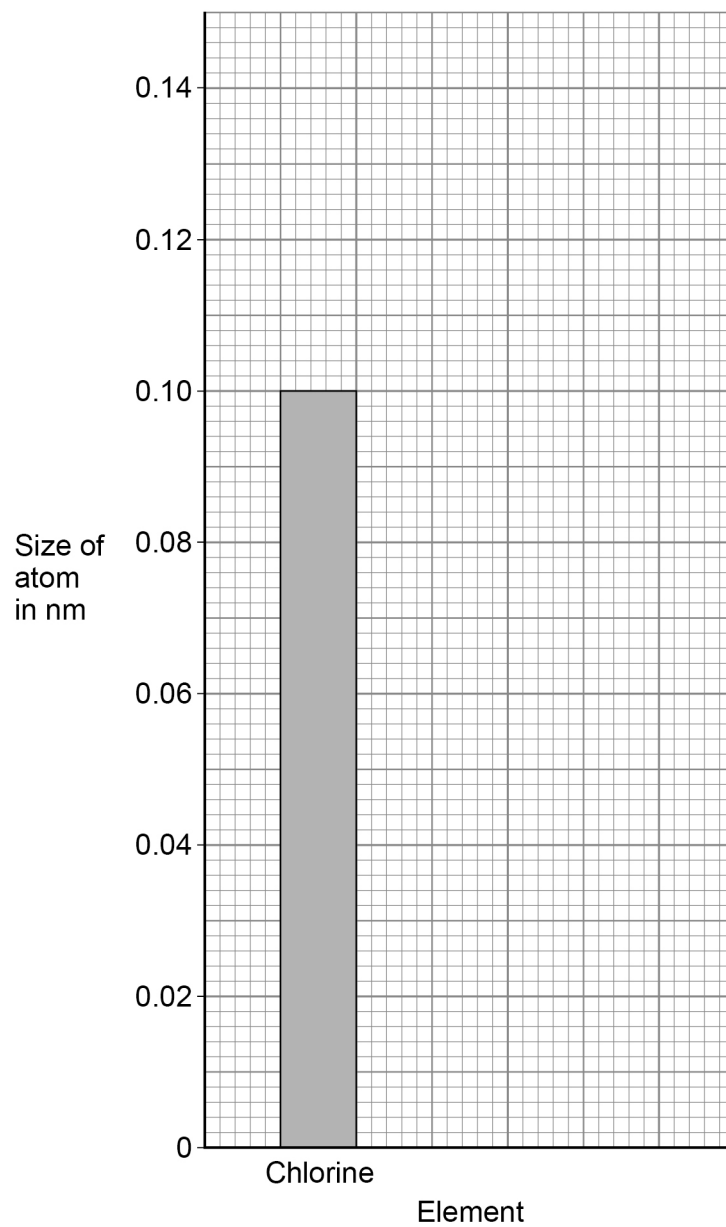
You should:

- plot the data from **Table 1** as a bar chart
- label each bar.

Chlorine has already been plotted.

[2 marks]



Figure 2

Question 2 continues on the next page

Turn over ►



0 2 . 3 Bromine can react in a displacement reaction.

Which aqueous solution reacts with bromine in a displacement reaction?

Use the periodic table.

[1 mark]

Tick (✓) **one** box.

Sodium bromide solution ☐

Sodium chloride solution ☐

Sodium fluoride solution ☐

Sodium iodide solution ☐

0 2 . 4 Bromine reacts with metals to form bromide ions.

What is the charge on a bromide ion?

[1 mark]

Tick (✓) **one** box.

-2 ☐ -1 ☐ +1 ☐ +2 ☐



Calcium is a Group 2 metal that reacts with chlorine.

0 2 . 5

How many electrons does a calcium atom lose in the reaction with chlorine?

[1 mark]

0 2 . 6

Calcium chloride is an ionic compound.

Describe the structure and bonding in an ionic compound.

[4 marks]

0 2 . 7

Table 2 shows some properties of four substances, **A**, **B**, **C** and **D**.

Table 2

Substance	Melting point in °C	Electrical conductivity when solid	Electrical conductivity when liquid
A	−39	Good	Good
B	1535	Good	Good
C	−117	Poor	Poor
D	1423	Poor	Good

Which substance is an ionic compound?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

D ☐



0	3
---	---

This question is about alkenes and polymers.

0	3	.	1
---	---	---	---

The general formula for the homologous series of alkenes is C_nH_{2n}

Determine the formula of an alkene with 7 carbon atoms.

[1 mark]

Formula = _____

0	3	.	2
---	---	---	---

Alkenes are unsaturated hydrocarbons.

Which part of the structure of alkenes makes them unsaturated?

[1 mark]



0 3 . 3 Dental polymers can be used to fill holes in teeth.

Table 3 shows some properties of polymers **E**, **F** and **G**.

Table 3

Polymer	Reactivity	Relative hardness	Solubility in water	Relative cost
E	Unreactive	Low	Soluble	Low
F	Reactive	High	Insoluble	Low
G	Unreactive	High	Insoluble	High

Evaluate the use of polymers **E**, **F** and **G** as dental polymers.

[4 marks]

Question 3 continues on the next page

Turn over ►



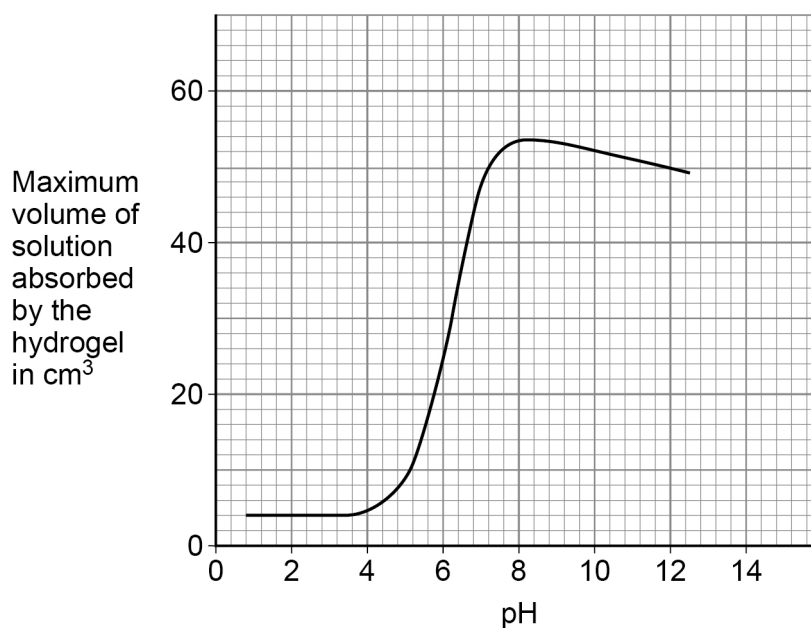
Hydrogels are a type of polymer that can absorb water from a solution and increase in volume.

Do not write
outside the
box

0 3 4

Figure 3 shows the effect of pH on the maximum volume of solution absorbed by a sample of hydrogel.

Figure 3



50 cm³ of a solution with a pH of 2.0 was added to the sample of the hydrogel.

Determine the percentage of the solution absorbed by the sample of the hydrogel.

[2 marks]

Percentage absorbed = _____ %

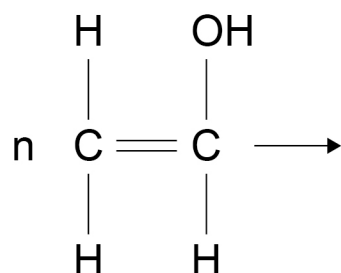


03.5

Monomer **X** is used to make a hydrogel polymer.

Complete the equation to show the formation of the polymer from monomer **X**.

[3 marks]



Do not write
outside the
box

11

Turn over for the next question

Turn over ►



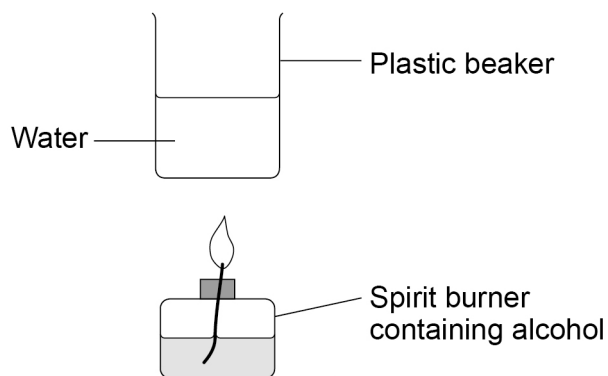
0	4
---	---

This question is about calorimetry.

A student investigated the amount of energy released by burning alcohols.

Figure 4 shows the apparatus.

Figure 4



0	4	.	1
---	---	---	---

The student should **not** have used a plastic beaker.

What material should the beaker be made from?

Give **one** reason for your choice.

[2 marks]

Material _____

Reason _____



Another student burned three different alcohols using the correct apparatus.

The student:

- calculated the experimental amount of energy released when 1 g of each alcohol was burned
- found out the expected amount of energy 1 g of each alcohol should have released when burned.

Table 4 shows the results.

Table 4

Name of alcohol	Number of carbon atoms in one molecule	Experimental amount of energy released when 1 g of alcohol was burned in kJ	Expected amount of energy 1 g of alcohol should have released when burned in kJ
Methanol	1	9.1	22.7
Ethanol	2	11.9	29.7
Butanol	4	14.3	36.2

0 4 . 2 The student concluded:

“The experimental amount of energy released is **not** directly proportional to the number of carbon atoms in one molecule.”

How do the results show that the relationship is **not** directly proportional?

Use **Table 4**.

[1 mark]

0 4 . 3 Suggest **one** reason why the experimental results are lower than the expected values.

Use **Table 4**.

[1 mark]

Question 4 continues on the next page

Turn over ►



0 4 . 4

A student heated 100 g of water by burning an alcohol in a spirit burner.

The student measured the mass of the spirit burner and the temperature of the water:

- before burning the alcohol
- after burning the alcohol.

Table 5 shows the results.

Table 5

	Mass of spirit burner in grams	Temperature of water in °C
Before burning	80.52	20.2
After burning	80.16	35.5

Calculate the amount of energy released when 1 g of the alcohol is burned.

Give your answer in kilojoules per gram (kJ/g).

Use **Table 5** and the equation: $Q = mc\Delta T$

Where:

- Q = energy released in J
- m = mass of water in g
- c = specific heat capacity = $4.2 \text{ J/g } ^\circ\text{C}$
- ΔT = temperature change in $^\circ\text{C}$.

[5 marks]

Energy released = _____ kJ/g



0 5

This question is about the extraction of metals.

0 5 . 1

Gold is found in the Earth as the metal itself.

Give **one** reason why.**[1 mark]**

0 5 . 2

Phytomining uses plants to absorb copper compounds from low-grade ores.

Table 6 shows some information about plants **A**, **B** and **C**, used in phytomining.**Table 6**

Plant	Metal ions extracted from the plant	Maximum mass of metal obtained from 1 kg of plant in mg	Maximum mass of plant that grows in 100 m ² area of land in kg
A	Cu ²⁺ Ni ²⁺	14 35	9.2
B	Cu ²⁺	10	13.0
C	Cu ²⁺	53	2.0

Evaluate the use of plants **A**, **B** and **C** for absorbing copper compounds by phytomining from a 100 m² area of land.

Include calculations in your answer.

[4 marks]

Question 5 continues on the next page**Turn over ►**

Bioleaching uses bacteria to produce copper compounds from low-grade copper ores.

This is the process used.

- 1 Break the copper ore into small pieces.
- 2 Add sulfuric acid and bacteria to the copper ore.
- 3 Leave to form a leachate solution.
- 4 Extract copper from the leachate solution.

0 5 . 3

Suggest why the copper ore is broken into small pieces in **step 1**.

[1 mark]

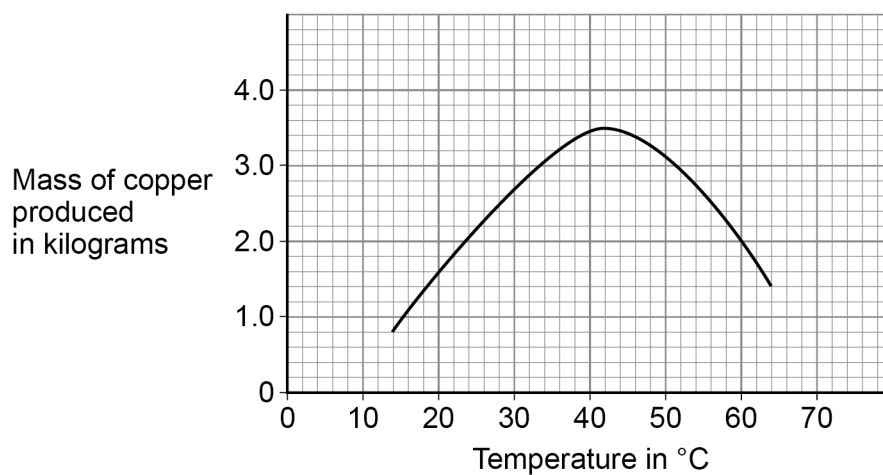


0 5 . 4

The mass of copper produced by bioleaching is affected by the temperature of the leachate solution.

Figure 5 shows the effect of temperature on the mass of copper produced.

Figure 5



Describe the trends shown in **Figure 5**.

Use data from **Figure 5**.

[3 marks]

Question 5 continues on the next page

Turn over ►

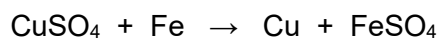


0 5 . 5

The leachate solution contains copper sulfate.

Copper can be produced from the leachate solution using scrap iron.

The equation for the reaction is:



381 g of copper is produced from the leachate solution.

Calculate the minimum mass of iron required to produce 381 g of copper.

Relative atomic masses (A_r): Cu = 63.5 Fe = 56

[3 marks]

Minimum mass of iron = _____ g

0 5 . 6

Copper is used in electrical wiring.

Explain why copper conducts electricity.

[2 marks]



Turn over for the next question

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Turn over ►



0 6

This question is about the reactions of metals with acid.

A student reacted magnesium with hydrochloric acid.

The student made the following observations:

- bubbles of gas were produced quickly
- the temperature of the solution increased by 12 °C.

0 6**. 1**

The student then reacted zinc with hydrochloric acid.

Give **two** ways in which the observations would be different from the reaction of magnesium with hydrochloric acid.

[2 marks]

1 _____

2 _____

The reaction of magnesium with hydrochloric acid produces a magnesium salt and hydrogen gas.

0 6**. 2**

Name the salt produced when magnesium reacts with hydrochloric acid.

[1 mark]

0 6**. 3**

Describe the test for hydrogen gas.

Give the result if hydrogen gas is present.

[2 marks]

Test _____

Result _____



0 6 . 4

Give **one** reason why reacting potassium with acid is **not** suitable for making a potassium salt.

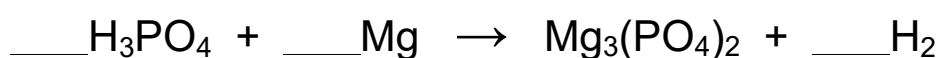
[1 mark]

0 6 . 5

Magnesium reacts with phosphoric acid (H_3PO_4).

Balance the equation for the reaction.

[1 mark]



0 6 . 6

The rate of a reaction that produces a gas can be determined by measuring the loss in mass.

Suggest **one** reason why measuring the loss in mass is **not** a suitable method for reactions that produce hydrogen gas (H_2).

[1 mark]

8

Turn over for the next question

Turn over ►



0 7

This question is about identifying ions.

0 7 . 1

Metal ions can be identified using flame tests.

Draw **one** line from each metal ion to the correct flame colour for that metal ion.**[2 marks]**

Metal ion	Flame colour
	Crimson
	Green
Barium	Lilac
Potassium	Red
	Yellow



Compounds can be identified using reactions in aqueous solution.

The compounds are:

- aluminium chloride
- calcium chloride
- magnesium bromide
- copper(II) bromide.

Plan a method to identify the compounds using reactions in aqueous solution.

Your method should include:

- the reagents used
- the observations made.

Do **not** refer to flame tests.

[6 marks]

[illegible]

8

Turn over for the next question

Turn over ►



0 8

This question is about precipitation.

Table 7 shows some information about the solubility of ionic compounds.

Table 7

Type of ionic compound	Solubility
Carbonates	None are soluble except Group 1 carbonates
Chlorides	All are soluble except lead chloride and silver chloride
Nitrates	All are soluble
Sulfates	All are soluble except barium sulfate and lead sulfate

A student mixed a solution containing lead ions with a solution containing chloride ions.

The reaction produced a precipitate of lead chloride.

0 8**1**

Name **two** solutions that can be mixed to produce a precipitate of lead chloride.

Use **Table 7**.

[2 marks]

1 _____

2 _____



0 8 . 2

Describe a method to find the mass of the lead chloride precipitate produced after the solutions have been mixed.

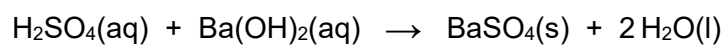
[4 marks]

Question 8 continues on the next page

Turn over ►

Adding barium hydroxide solution to sulfuric acid produces a precipitate of barium sulfate.

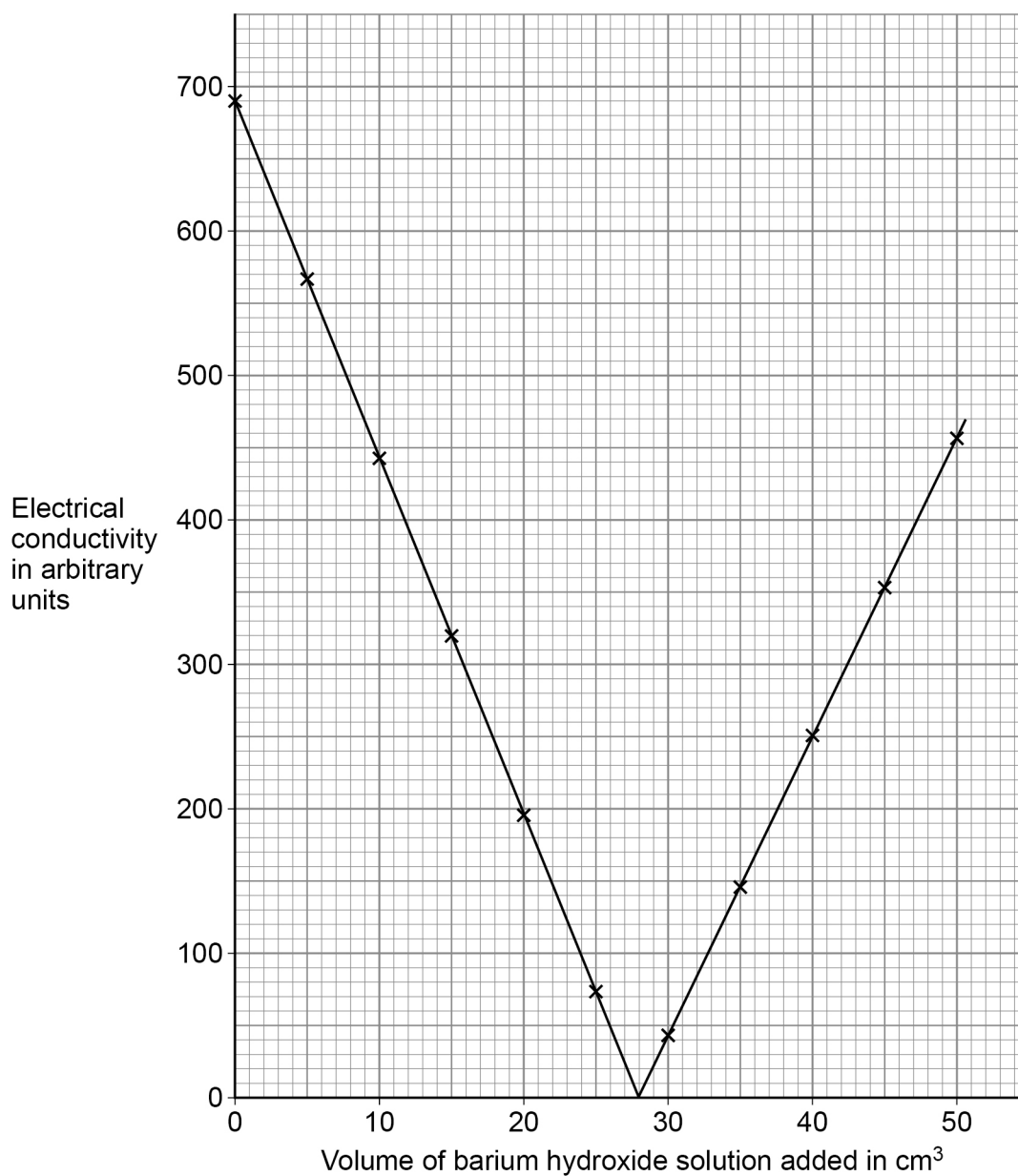
The equation for the reaction is:



A student measured the electrical conductivity of the mixture as barium hydroxide solution was added to 25 cm³ of sulfuric acid.

Figure 6 shows the results.

Figure 6



0 8 . 3

What volume of barium hydroxide solution produced an electrical conductivity of zero?

Use **Figure 6**.

[1 mark]

Volume of barium hydroxide solution = _____ cm³

0 8 . 4

Suggest **one** reason why the electrical conductivity becomes zero.

[1 mark]

0 8 . 5

Calculate the mass of barium hydroxide (Ba(OH)₂) needed to make 50 cm³ of 0.80 mol/dm³ solution.Relative formula mass (*M_r*) of Ba(OH)₂ = 171

[4 marks]

Mass of barium hydroxide = _____ g

12

Turn over for the next question

Turn over ►



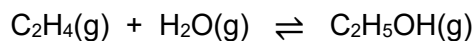
0 9

This question is about chemical reactions.

0 9 . 1

A scientist produced ethanol by heating 1.0 kg of ethene with excess steam in a container.

The equation for the reaction is:



The scientist calculated that 1.0 kg of ethene should produce 1.6 kg of ethanol.

At the end of the experiment the scientist collected 80 g of ethanol.

The scientist also noticed an unexpected white solid in the container.

Give **three** reasons why the scientist did not obtain the calculated mass of product.

[3 marks]

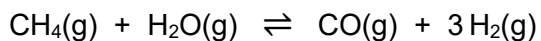
1 _____

2 _____

3 _____

Hydrogen gas can be produced from methane.

The equation for the reaction is:



0 9 . 2

Explain the effect on the equilibrium position of increasing the pressure.

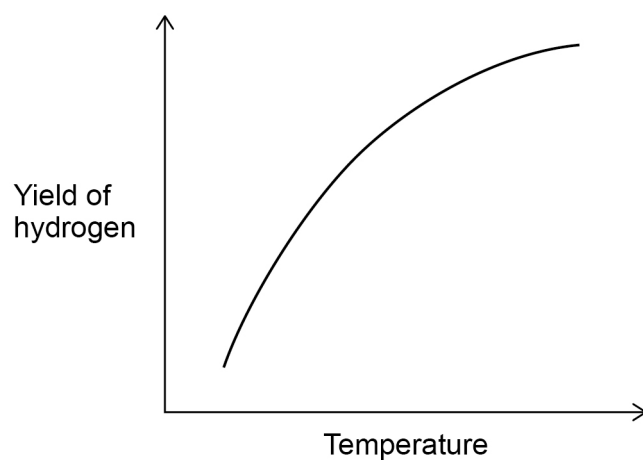
[2 marks]



0 9 . 3

Figure 7 shows the change in the yield of hydrogen as the temperature of the equilibrium mixture is increased.

Figure 7



Explain the effect on the equilibrium position of increasing the temperature.

[3 marks]

8

END OF QUESTIONS



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



*Do not write
outside the
box*

[illegible]

*Do not write
outside the
box*

[illegible]

[illegible]

There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Copyright information

For confidentiality purposes, all acknowledgements of third-party copyright material are published in a separate booklet. This booklet is published after each live examination series and is available for free download from www.oxfordaqaexams.org.uk.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and Oxford International AQA Examinations will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team.

Copyright © 2022 Oxford International AQA Examinations and its licensors. All rights reserved.

